

67% of the zeroes are on the line.

Theorem.
$$\sum_{\substack{\zeta(\rho)=0, \Re \rho=\frac{1}{2}, \\ T \leq \Im \rho < 2T}} 1 \geq \left(\underbrace{\frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}}_{= 67.25\ldots\%} - o_{T \rightarrow \infty}(1) \right) \cdot \sum_{\substack{\zeta(\rho)=0, \\ T \leq \Im \rho < 2T}} \text{ord}_{\rho} \zeta.$$

We are going to choose a good finite-dimensional space of functions and restrict the pairing induced by the explicit formula to that space. That restriction will have the property that zeroes on the line contribute a positive line aka $(1, 0)$ part, and pairs of zeroes off the line contribute a hyperbolic plane aka $(1, 1)$ part. Then we will write down an inequality lower bounding the rank of the part attributed to the zeroes on the line in terms of information controlled by the 2-norm of the full form as well as traces of these on- and off-the-line constituents. And then we will manage to calculate everything.

Proof of Theorem assuming all claims and lemmas below.

$$\begin{aligned} \#|\text{on}_{[T, 2T]}| + O\left(T^{\delta} \log T\right) &= \#|\text{on}_{N_{T^{\delta}}([T, 2T])}| \geq \text{rank } W_T^{\text{on}} & (v_{\rho} \cdot v_{\rho}^t \geq 0 \text{ if } \Re \rho = \tfrac{1}{2}) \\ &\geq 2 \text{tr } W_T^{\text{on}} + 4 \text{tr } W_T^{\text{off}} - 4n_+ \left(W_T^{\text{off}}\right) - \|W_T\|_{\text{HS}}^2 & (\text{Lemma 3.4}) \\ &= 4 \text{tr } W_T - 2 \left(\text{tr } W_T^{\text{on}} + 2n_+ \left(W_T^{\text{off}}\right) \right) - \|W_T\|_{\text{HS}}^2 & (W_T = W_T^{\text{on}} + W_T^{\text{off}}) \\ &\geq 4 \text{tr } W_T - 2N_{\text{all}}(T) - \|W_T\|_{\text{HS}}^2 - O\left(T^{\delta} \log T\right) & (\text{Claim 2.3}) \\ &= \left(4 - 2 - \frac{1}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} + o_{\chi; T \rightarrow \infty}(1) \right) \cdot N_{\text{all}}(T) & (\text{Lemmas 3.2 \& 3.3}) \\ &= \left(\frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} + o_{\chi; T \rightarrow \infty}(1) \right) \cdot N_{\text{all}}(T), \end{aligned}$$

as desired. $\square$

1 Notation.

Let $T \gg 1$. Write, for an interval $I \subseteq \mathbb{R}$, $N_{\text{all}}(I) := \sum_{\substack{\zeta(\rho)=0, \\ \Im \rho \in I}} \text{ord}_{\rho} \zeta$ and $N_{\text{all}}(T) := N_{\text{all}}([T, 2T])$ — thus $N_{\text{all}}(T) = \frac{T}{2\pi} \log\left(\frac{2T}{\pi}\right) - \frac{T}{2\pi} + O(\log T)$.¹ Let $\delta := 10^{-10}$.² Let $\psi(x) := \cos(\sqrt{2}x) \cdot \mathbf{1}_{|x| \leq \frac{1}{2}}$.³ Let $\chi \in C^{\infty}(\mathbb{R})$ be such that $\chi|_{(-\infty, 0]} = 0$, $\chi|_{[1, \infty)} = 1$, and χ is nondecreasing. Let $\varphi_T(x) := \chi\left(\frac{N_{\text{all}}(T)}{2T} + x\right) \cdot \chi\left(\frac{N_{\text{all}}(T)}{2T} - x\right) \cdot \sqrt{\cos\left(\frac{\sqrt{2}T}{N_{\text{all}}(T)} \cdot x\right)}$. Then $\varphi_T \in C_c^{\infty}(\mathbb{R})$, is even, $0 \leq \varphi_T \leq 1$, and $\text{supp } \varphi_T \subseteq \left[-\frac{N_{\text{all}}(T)}{2T}, \frac{N_{\text{all}}(T)}{2T}\right]$. Applying Paley-Wiener to continue $\widehat{\varphi}_T$, $|\widehat{\varphi}_T(z)| \ll_{\chi} e^{\pi \cdot |\Im z| \cdot \frac{N_{\text{all}}(T)}{T}} \cdot \min(\log T, |z|^{-1}, |z|^{-2})$ by integrating by parts 0, 1, or 2 times. Let $\mu(\tau) := \frac{1}{2\pi} \Re\left(\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{i\tau}{2}\right)\right) - \frac{\log \pi}{2\pi}$, $\lambda_T(\tau) := -\frac{1}{\pi} \sum_{n \leq T} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n)$, $\pi_T(\tau) := \frac{1}{\pi} \Re\left(\frac{T^{\frac{1}{2} + i\tau}}{\frac{1}{2} + i\tau}\right)$, and $\nu_T := \mu + \lambda_T + \pi_T$. Let, for $\rho \in \mathbb{C}$, $\gamma_{\rho} := \frac{\rho - \frac{1}{2}}{i}$.

Theorem (Guinand-Weil). Let $F \in C_c^{\infty}(\mathbb{R})$. Then:

$$\sum_{\rho} \text{ord}_{\rho} \zeta \cdot \widehat{F}(\gamma_{\rho}) = \widehat{F}\left(\frac{i}{2}\right) + \widehat{F}\left(-\frac{i}{2}\right) + \int_{\mathbb{R}} d\tau \widehat{F}(\tau) \cdot \mu(\tau) - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \left(F\left(\frac{\log n}{2\pi}\right) + F\left(-\frac{\log n}{2\pi}\right) \right).$$

Thus if $\text{supp } F \subseteq \left[-\frac{N_{\text{all}}(T)}{T}, \frac{N_{\text{all}}(T)}{T}\right] \subsetneq \left[-\frac{\log T}{2\pi}, \frac{\log T}{2\pi}\right]$ then $\sum_{\rho} \text{ord}_{\rho} \zeta \cdot \widehat{F}(\gamma_{\rho}) = \int_{\mathbb{R}} d\tau \widehat{F}(\tau) \cdot \nu_T(\tau)$. Let now $v_{\rho} := \left(\widehat{\varphi}_T\left(\gamma_{\rho} - T - \frac{T}{N_{\text{all}}(T)} \cdot k\right)\right)_{k=0}^{N_{\text{all}}(T)-1} \in \mathbb{C}^{N_{\text{all}}(T)}$. Let $W_T := \frac{T}{N_{\text{all}}(T) \cdot \int_{\mathbb{R}} \varphi_T^2} \sum_{\Im \rho \in N_{T^{\delta}}([T, 2T])} \text{ord}_{\rho} \zeta \cdot v_{\rho} \cdot v_{\rho}^t$. It

¹(Riemann-von Mangoldt aka integrate $\frac{\zeta'}{\zeta}$ in a rectangle with top and bottom to infinity, right at $\Re s = 1 + \delta$, and left and right matched up by the functional equation.)

²Its use indicates unoptimized but easy parts.

³This is the minimizer of $\psi \mapsto \frac{\int_{\mathbb{R}} \psi^2 + \int_{\mathbb{R}} du \int_{\mathbb{R}} dv \psi(u) \psi(v) \cdot |u-v|}{\left(\int_{\mathbb{R}} \psi\right)^2}$ and the provenance of the $\frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}$.

should be clear that roots on the line contribute positive squares in the sum, whereas pairs $(\rho, 1 - \bar{\rho})$ of roots off the line contribute hyperbolic planes, **and extracting counts based on the signature of this form via first- and second-order information is the entire point**. Because the functional equation matches roots and multiplicities off the line W_T is real symmetric. Let, for an interval $I \subseteq \mathbb{R}$, $\text{on}_I := \{\rho : \zeta(\rho) = 0, \Re \rho = \frac{1}{2}, \Im \rho \in I\}$, $\text{off}_I := \{\rho : \zeta(\rho) = 0, \Re \rho \neq \frac{1}{2}, \Im \rho \in I\}$, and $\text{far}_I := \{\rho : \zeta(\rho) = 0, \Im \rho \notin I\}$. Thus by Riemann-von Mangoldt $\#|\text{on}_{N_{T^\delta}([T, 2T])}| = \#|\text{on}_{[T, 2T]}| + O(T^\delta \log T)$. Also $\sum_{\rho \in \text{on}_I} \text{ord}_\rho \zeta + \#|\text{off}_I| \leq N_{\text{all}}(I)$.

Let $W_T^{\text{on}} := \frac{T}{N_{\text{all}}(T) \cdot \int_{\mathbb{R}} \varphi_T^2} \sum_{\rho \in \text{on}_{N_{T^\delta}([T, 2T])}} \text{ord}_\rho \zeta \cdot v_\rho \cdot v_\rho^t$, $W_T^{\text{off}} := \frac{T}{N_{\text{all}}(T) \cdot \int_{\mathbb{R}} \varphi_T^2} \sum_{\rho \in \text{off}_{N_{T^\delta}([T, 2T])}} \text{ord}_\rho \zeta \cdot v_\rho \cdot v_\rho^t$, so that $W_T = W_T^{\text{on}} + W_T^{\text{off}}$. Hence W_T^{on} is positive semidefinite and $\text{rank } W_T^{\text{on}} \leq \#|\text{on}_{N_{T^\delta}([T, 2T])}|$. We write the signature of a quadratic form Q as $(n_+(Q), n_-(Q))$.

Let $E_{\text{far}} := \frac{T}{N_{\text{all}}(T) \cdot \int_{\mathbb{R}} \varphi_T^2} \sum_{\rho \in \text{far}_{N_{T^\delta}([T, 2T])}} \text{ord}_\rho \zeta \cdot v_\rho \cdot v_\rho^t$.

Theorem 1.1 (Montgomery-Vaughan). *Let $x_n \in \mathbb{C}$ and $y_n \in \mathbb{R}$ be such that for each m there is a $\sigma_m > 0$ such that for all n either $y_m = y_n$ or else $|y_m - y_n| \geq \sigma_m$. Then:*

$$\left| \sum_{m \neq n} \frac{x_m \cdot \bar{x}_n}{y_m - y_n} \right| \ll \sum_m |x_m|^2 \cdot \sigma_m^{-1}.$$

2 Claims.

Claim 2.1. *Let $z, z' \in \mathbb{C}$. Then:*

$$\sum_{k \in \mathbb{Z}} \widehat{\varphi}_T \left(z - T - \frac{kT}{N_{\text{all}}(T)} \right) \cdot \widehat{\varphi}_T \left(z' - T - \frac{kT}{N_{\text{all}}(T)} \right) = \frac{N_{\text{all}}(T)}{T} \cdot \widehat{\varphi}_T^2(z - z').$$

Proof of Claim 2.1. Apply Poisson summation and simply compute the Fourier transform — the point being that $\text{supp } \varphi_T + \text{supp } \varphi_T \subseteq \left[-\frac{N_{\text{all}}(T)}{T}, \frac{N_{\text{all}}(T)}{T} \right]$ and so all but the zero mode of the relevant convolution vanishes. □

Claim 2.2. *For $0 \leq k, \ell < N_{\text{all}}(T)$,*

$$W_{k, \ell} = \frac{T}{N_{\text{all}}(T) \cdot \int_{\mathbb{R}} \varphi_T^2} \left(\int_{\mathbb{R}} d\tau \widehat{\varphi}_T \left(\tau - T - \frac{T}{N_{\text{all}}(T)} k \right) \cdot \widehat{\varphi}_T \left(\tau - T - \frac{T}{N_{\text{all}}(T)} \ell \right) \cdot \nu_T(\tau) \right. \\ \left. - \sum_{\rho \in \text{far}_{N_{T^\delta}([T, 2T])}} \text{ord}_\rho \zeta \cdot \widehat{\varphi}_T \left(\gamma_\rho - T - \frac{T}{N_{\text{all}}(T)} k \right) \cdot \widehat{\varphi}_T \left(\gamma_\rho - T - \frac{T}{N_{\text{all}}(T)} \ell \right) \right).$$

Proof of Claim 2.2. The claim is equivalently a calculation of

$$\sum_{\rho} \text{ord}_\rho \zeta \cdot \widehat{\varphi}_T \left(\gamma_\rho - T - \frac{T}{N_{\text{all}}(T)} \cdot k \right) \cdot \widehat{\varphi}_T \left(\gamma_\rho - T - \frac{T}{N_{\text{all}}(T)} \cdot \ell \right),$$

and per the remark after our statement of Guinand-Weil it suffices to check that the relevant convolution of two φ_T 's has support in $\left[-\frac{N_{\text{all}}(T)}{T}, \frac{N_{\text{all}}(T)}{T} \right]$, which it does. □

Claim 2.3. $\text{tr } W_T^{\text{on}} \leq \sum_{\rho \in \text{on}_{N_{T^\delta}([T, 2T])}} \text{ord}_\rho \zeta$ and $n_+(W_T^{\text{off}}) \leq \frac{1}{2} \cdot \#|\text{off}_{N_{T^\delta}([T, 2T])}|$.

Proof of Claim 2.3. For $\rho \in \text{on}_{N_{T^\delta}([T, 2T])}$ one has $\gamma_\rho \in \mathbb{R}$, so

$$\text{tr } (v_\rho \cdot v_\rho^t) = \|v_\rho\|_2^2 = \sum_{k=0}^{N_{\text{all}}(T)-1} \widehat{\varphi}_T \left(\gamma_\rho - T - \frac{T}{N_{\text{all}}(T)} \cdot k \right)^2 \leq \frac{N_{\text{all}}(T)}{T} \cdot \int_{\mathbb{R}} \varphi_T^2$$

by Claim 2.1 at $z = z' = \gamma_\rho$. Summing gives the trace bound.

For the rest, pair off $\rho \in \text{off}_{N_{T^\delta}([T, 2T])}$ as $(\rho, 1 - \bar{\rho})$ via the functional equation. Because $\gamma_{1-\bar{\rho}} = \overline{\gamma_\rho}$ it follows that $v_{1-\bar{\rho}} = \overline{v_\rho}$ (Schwarz reflection), whence the desired hyperbolic plane i.e. $v_\rho \cdot v_\rho^t + \overline{v_\rho} \cdot \overline{v_\rho^t}$. It follows that the relevant quadratic form is a pullback from $(1, 1)^{\oplus \#|\text{pairs}|}$ and the bound follows from comparing signatures. □

3 Lemmas.

Lemma 3.1. $|\operatorname{tr} E_{\text{far}}|, \|E_{\text{far}}\|_{\text{HS}} \ll_{\chi} T^{\frac{1}{2}-2\delta}.$

Proof of Lemma 3.1. Run the same calculation as in Claim 2.3 and apply the Paley-Wiener integration by parts bound and sum (the square-root coming from $|\Im \gamma_{\rho}| < \frac{1}{2}$ a priori). $\square$

Lemma 3.2. $\operatorname{tr} W_T = (1 + o_{\chi; T \rightarrow \infty}(1)) \cdot N_{\text{all}}(T).$

Proof of Lemma 3.2. Again apply Paley-Wiener to control the contribution of $k \notin [0, N_{\text{all}}(T) - 1]$ by $\ll_{\chi} \sqrt{T} \cdot \frac{N_{\text{all}}(T)}{T} \cdot \operatorname{dist}(\Im \rho, \{T, 2T\})^{-3}$ (and similarly for the $\ll T^{\delta}$ many ρ with $\Im \rho$ within T^{δ} of the interval endpoints). $\square$

Lemma 3.3. $\|W_T\|_{\text{HS}}^2 = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} + o_{\chi; T \rightarrow \infty}(1)\right) \cdot N_{\text{all}}(T).$

Proof of Lemma 3.3. The Hilbert-Schmidt norm is, after completing the relevant k -sum and localizing the relevant τ and τ' integrals to $[T, 2T]$, an integral of a kernel formed from a convolution of φ_T with itself against a product of ν_T with itself. ν_T has three terms: π_T , corresponding to the pole of ζ , is irrelevant here. The diagonal terms of the $\lambda_T \cdot \lambda_T$ product will contribute to the main term, and for the off-diagonal terms we will use Montgomery-Vaughan-Hilbert. $\mu \cdot \mu$ also contributes, and this is where we see the shape of the variational problem that ψ solves. All other cross-terms cancel (e.g. because of the cosine oscillation in λ_T).

Let $K(\tau, \tau') := \sum_{0 \leq k < N_{\text{all}}(T)} \widehat{\varphi}_T\left(\tau - T - \frac{T}{N_{\text{all}}(T)} \cdot k\right) \cdot \widehat{\varphi}_T\left(\tau' - T - \frac{T}{N_{\text{all}}(T)} \cdot k\right)$ and $\alpha_k := T + \frac{T}{N_{\text{all}}(T)} \cdot k$.

By definition of E_{far} , $\left(\frac{N_{\text{all}}(T)}{T} \cdot \int_{\mathbb{R}} \varphi_T^2\right)^2 \cdot \|W_T + E_{\text{far}}\|_{\text{HS}}^2 = \int_{\mathbb{R}} d\tau \int_{\mathbb{R}} d\tau' K(\tau, \tau')^2 \cdot \nu_T(\tau) \cdot \nu_T(\tau')$. We will calculate the right-hand side and then use Lemma 3.1 to remove the E_{far} from the left-hand side. (Note that we expect the main term to be of order $T \cdot (\log T)^5$.)

First let us execute the Poisson summation à la Claim 2.1 and also localize the integrals to $[T, 2T]$.

Throughout the error terms below we freely write $\log T$ for $\frac{N_{\text{all}}(T)}{T}$ (and hence $\log \log T$ for $\log \frac{N_{\text{all}}(T)}{T}$); only main terms need the exact ratio. Record once: for real τ, τ' one has $|K(\tau, \tau')| \leq \frac{N_{\text{all}}(T)}{T} \cdot \int_{\mathbb{R}} \varphi_T^2 \ll \log^2 T$ (Cauchy-Schwarz + Claim 2.1 at $z = z' = \tau$ resp. τ'); $|\widehat{\varphi}_T^2(s)| \leq \int_{\mathbb{R}} \varphi_T^2 \ll \log T$; and $\int_{|s| \geq D} ds |\widehat{\varphi}_T(s)| \ll_{\chi} \min(\log \log T, \frac{1}{D})$ (integrate the Paley-Wiener min).

Claim. $\int_{\mathbb{R}} d\tau \int_{\mathbb{R}} d\tau' K(\tau, \tau')^2 \cdot \nu_T(\tau) \cdot \nu_T(\tau') = \left(\frac{N_{\text{all}}(T)}{T}\right)^2 \cdot \int_T^{2T} d\tau \int_T^{2T} d\tau' \widehat{\varphi}_T^2(\tau - \tau')^2 \cdot \nu_T(\tau) \cdot \nu_T(\tau') + O_{\chi}(T \cdot \log^4 T \cdot \log \log T).$

Proof. $\square [T, 2T]^2$: by Claim 2.1, $K(\tau, \tau') - \frac{N_{\text{all}}(T)}{T} \cdot \widehat{\varphi}_T^2(\tau - \tau') = -\sum_{k \notin [0, N_{\text{all}}(T))} \widehat{\varphi}_T(\tau - \alpha_k) \cdot \widehat{\varphi}_T(\tau' - \alpha_k)$. Hence

$$\left|K(\tau, \tau')^2 - \left(\frac{N_{\text{all}}(T)}{T} \cdot \widehat{\varphi}_T^2(\tau - \tau')\right)^2\right| \ll \log^2 T \cdot \sum_{k \notin [0, N_{\text{all}}(T))} |\widehat{\varphi}_T(\tau - \alpha_k) \cdot \widehat{\varphi}_T(\tau' - \alpha_k)|.$$

Integrate against $|\nu_T(\tau) \cdot \nu_T(\tau')| \leq T$ (Chebyshev + Stirling: $|\mu| \ll \log T$, $|\lambda_T| \ll \sqrt{T}$, $|\pi_T| \ll T^{-\frac{1}{2}}$ on $[T, 2T]$) over $[T, 2T]^2$; the k -th summand factors as $\left(\int_T^{2T} d\tau |\widehat{\varphi}_T(\tau - \alpha_k)|\right)^2$. For $k \notin [0, N_{\text{all}}(T))$ one has $\alpha_k \notin [T, 2T]$, so $|\tau - \alpha_k| \geq \frac{T}{N_{\text{all}}(T)} \cdot \operatorname{dist}(k, [0, N_{\text{all}}(T)])$ for all $\tau \in [T, 2T]$; hence $\int_T^{2T} d\tau |\widehat{\varphi}_T(\tau - \alpha_k)| \ll_{\chi} \min\left(\log \log T, \frac{\log T}{\operatorname{dist}(k, [0, N_{\text{all}}(T)])}\right)$. Summing the squares:

$$\sum_{k \notin [0, N_{\text{all}}(T))} \min\left(\log \log T, \frac{\log T}{\operatorname{dist}(k, \dots)}\right)^2 \ll \log T \cdot \log \log T.$$

So this piece is $\ll_{\chi} T \cdot \log^3 T \cdot \log \log T$.

Off $[T, 2T]^2$: by symmetry $\leq 2 \int_{\tau \in \mathbb{R}} \int_{\tau' \notin [T, 2T]} \cdot$. Pull one $|K| \ll \log^2 T$ out: $K(\tau, \tau')^2 \ll \log^2 T \cdot \sum_{0 \leq k < N_{\text{all}}(T)} |\widehat{\varphi}_T(\tau - \alpha_k) \cdot \widehat{\varphi}_T(\tau' - \alpha_k)|$. Factor the double integral:

$$\ll \log^2 T \cdot \sum_{0 \leq k < N_{\text{all}}(T)} \left(\int_{\mathbb{R}} d\tau |\widehat{\varphi}_T(\tau - \alpha_k)| \cdot |\nu_T|\right) \cdot \left(\int_{\mathbb{R} \setminus [T, 2T]} d\tau' |\widehat{\varphi}_T(\tau' - \alpha_k)| \cdot |\nu_T|\right).$$

The first factor: $|\nu_T| \ll \sqrt{T}$ on $|\tau| \leq T^2$, so $\leq \sqrt{T} \cdot \int_{\mathbb{R}} |\widehat{\varphi}_T| \ll \sqrt{T} \cdot \log \log T$ (the $|\tau| > T^2$ tail is negligible via the $|\tau|^{-2}$ decay). The second factor: $\alpha_k \in [T, 2T]$, so for $\tau' \notin [T, 2T]$ one has $|\tau' - \alpha_k| \geq \frac{T}{N_{\text{all}}(T)} \cdot \min(k, N_{\text{all}}(T) - k)$; hence $\ll \sqrt{T} \cdot \min\left(\log \log T, \frac{\log T}{\min(k, N_{\text{all}}(T) - k)}\right)$. Summing:

$$\sum_{0 \leq k < N_{\text{all}}(T)} \min\left(\log \log T, \frac{\log T}{\min(k, N_{\text{all}}(T) - k)}\right) \ll \log T \cdot \log N_{\text{all}}(T) \ll \log^2 T.$$

So this piece is $\ll_{\chi} \log^2 T \cdot \sqrt{T} \cdot \log \log T \cdot \sqrt{T} \cdot \log^2 T = T \cdot \log^4 T \cdot \log \log T$. $\square$

Now let us split $\nu_T \cdot \nu_T = (\mu + \lambda_T + \pi_T)^2$. In each piece the error is at least one $\log T$ below the main term $\asymp T \cdot \log^3 T$.

Claim. $\int_T^{2T} d\tau \int_T^{2T} d\tau' \widehat{\varphi}_T^2(\tau - \tau')^2 \cdot \mu(\tau) \cdot \mu(\tau') = T \cdot \left(\frac{N_{\text{all}}(T)}{T}\right)^2 \cdot \int_{\mathbb{R}} \varphi_T^4 + O_{\chi}(T \cdot \log^2 T)$.

Proof. Write $\mu(\tau) = \frac{N_{\text{all}}(T)}{T} + O(1)$ on $[T, 2T]$ (Stirling: $\mu(\tau) = \frac{1}{2\pi} \log \frac{\tau}{2\pi} + O(\tau^{-1})$); and $\frac{N_{\text{all}}(T)}{T}$ is the average of μ over $[T, 2T]$ up to $O(\frac{\log T}{T})$ by Riemann-von Mangoldt and $\mu(\tau) - \mu(\tau') \ll \frac{|\tau - \tau'|}{T}$ (since $\mu' \ll |\tau|^{-1}$). Freeze $\mu(\tau') \rightarrow \mu(\tau)$: cost $\ll \int_T^{2T} d\tau |\mu(\tau)| \cdot \frac{1}{T} \cdot \int_{\mathbb{R}} ds \widehat{\varphi}_T^2(s)^2 \cdot |s| \ll \log T \cdot \log \log T$ (via $\int_{\mathbb{R}} ds \widehat{\varphi}_T^2(s)^2 \cdot |s| \ll_{\chi} \log \log T$). Extend $\tau' \in [T, 2T] \rightarrow \tau' \in \mathbb{R}$: cost $\ll \int_T^{2T} d\tau \mu(\tau)^2 \cdot \int_{|s| \geq \text{dist}(\tau, \{T, 2T\})} ds \widehat{\varphi}_T^2(s)^2 \ll_{\chi} \log^2 T \cdot (\log T)^{2/3}$. Then $\int_{\mathbb{R}} ds \widehat{\varphi}_T^2(s)^2 = \int_{\mathbb{R}} \varphi_T^4$ (Plancherel), and $\int_T^{2T} \mu^2 = T \cdot \left(\frac{N_{\text{all}}(T)}{T}\right)^2 + O(N_{\text{all}}(T))$ (from $\mu = \frac{N_{\text{all}}(T)}{T} + O(1)$ squared). $\square$

Claim. $\int_T^{2T} d\tau \int_T^{2T} d\tau' \widehat{\varphi}_T^2(\tau - \tau')^2 \cdot \lambda_T(\tau) \cdot \lambda_T(\tau') = 2T \cdot \int_0^{N_{\text{all}}(T)/T} dy y \cdot (\varphi_T^2 * \varphi_T^2)(y) + O_{\chi}(T \cdot \log^2 T)$.

Proof. Extend $\tau' \rightarrow \mathbb{R}$: cost $\ll |\lambda_T|_{\infty}^2 \cdot \int_{\tau \in [T, 2T], \tau' \notin [T, 2T]} \widehat{\varphi}_T^2(\tau - \tau')^2 \ll_{\chi} T \cdot (\log T)^{2/3}$ (via $|\lambda_T| \ll \sqrt{T}$ and $\int_{\tau \in [T, 2T], \tau' \notin [T, 2T]} \widehat{\varphi}_T^2(\tau - \tau')^2 = \int_0^T dD \int_{|s| \geq D} ds \widehat{\varphi}_T^2(s)^2 \ll_{\chi} (\log T)^{2/3}$).

Inner integral. Since $\int_{\mathbb{R}} ds \widehat{\varphi}_T^2(s)^2 \cdot e(sy) = (\varphi_T^2 * \varphi_T^2)(y)$ and $\widehat{\varphi}_T^2$ is even,

$$\int_{\mathbb{R}} d\tau' \widehat{\varphi}_T^2(\tau - \tau')^2 \cdot \lambda_T(\tau') = -\frac{1}{\pi} \sum_m \frac{\Lambda(m)}{\sqrt{m}} \cdot (\varphi_T^2 * \varphi_T^2)\left(\frac{\log m}{2\pi}\right) \cdot \cos(\tau \log m).$$

Multiply by $\lambda_T(\tau)$ and integrate over $\tau \in [T, 2T]$; write $\cos A \cdot \cos B = \frac{1}{2}(\cos(A+B) + \cos(A-B))$ with $A := \tau \log n$, $B := \tau \log m$.

Diagonal $n = m$. $\int_T^{2T} d\tau \cos^2(\tau \log n) = \frac{T}{2} + O\left(\frac{1}{\log n}\right)$, so this piece is $\frac{T}{2\pi^2} \sum_n \frac{\Lambda(n)^2}{n} \cdot (\varphi_T^2 * \varphi_T^2)\left(\frac{\log n}{2\pi}\right) + O_{\chi}(\log^2 T)$. *Partial summation:* with $A(t) := \sum_{n \leq t} \frac{\Lambda(n)^2}{n} = \frac{1}{2} \log^2 t + O(\log t)$ and $g(y) := (\varphi_T^2 * \varphi_T^2)(y)$ (so $\text{supp } g \subseteq \left[-\frac{N_{\text{all}}(T)}{T}, \frac{N_{\text{all}}(T)}{T}\right]$ and $\|g'\|_{\infty} \leq \|(\varphi_T^2)'\|_1 \ll_{\chi} 1$),

$$\sum_n \frac{\Lambda(n)^2}{n} \cdot g\left(\frac{\log n}{2\pi}\right) = -\int_0^{N_{\text{all}}(T)/T} dy \left(\frac{1}{2}(2\pi y)^2 + O(2\pi y)\right) \cdot g'(y) = 4\pi^2 \int_0^{N_{\text{all}}(T)/T} dy y \cdot g(y) + O_{\chi}(\log^2 T)$$

(the boundary term $A(T) \cdot g\left(\frac{\log T}{2\pi}\right) = 0$ since $\frac{\log T}{2\pi} > \frac{N_{\text{all}}(T)}{T}$; then integrate by parts once more). Times $\frac{T}{2\pi^2}$: $2T \int_0^{N_{\text{all}}(T)/T} dy y \cdot g(y) + O_{\chi}(T \cdot \log^2 T)$.

Off-diagonal $n \neq m$. $\int_T^{2T} d\tau \cos\left(\tau \log \frac{n}{m}\right) = \Im \left(\frac{(n/m)^{2iT} - (n/m)^{-iT}}{\log(n/m)} \right)$. Set $x_n := \frac{\Lambda(n)}{\sqrt{n}} \cdot n^{iT}$ and $x'_n := x_n \cdot g\left(\frac{\log n}{2\pi}\right)$ (respectively n^{2iT}): the sum is $\Im \sum_{n \neq m} \frac{x_n \cdot \overline{x'_m}}{\log n - \log m}$, and Montgomery-Vaughan with $y_n := \log n$, $\sigma_n > \frac{1}{n+1}$ gives (in bilinear form: the theorem reads $-CD \leq iH \leq CD$, so $\|D^{-\frac{1}{2}} \cdot iH \cdot D^{-\frac{1}{2}}\|_{\text{op}} \leq C$) $|\dots| \ll (\sum_{n \leq T} \Lambda(n)^2)^{\frac{1}{2}} \cdot \|g\|_{\infty} \cdot (\sum_{m \leq T} \Lambda(m)^2)^{\frac{1}{2}} \ll T \cdot \log^2 T$.

The $\cos(A+B)$ -half. $\left| \int_T^{2T} d\tau \cos(\tau \log nm) \right| \leq \frac{2}{\log 4}$; so this is $\ll \log T \cdot \left(\sum_n \frac{\Lambda(n)}{\sqrt{n}}\right) \cdot \left(\sum_m \frac{\Lambda(m)}{\sqrt{m}}\right) \ll T$. $\square$

Claim. $\int_T^{2T} d\tau \int_T^{2T} d\tau' \widehat{\varphi}_T^2(\tau - \tau')^2 \cdot (2\mu\lambda_T + 2\mu\pi_T + 2\lambda_T\pi_T + \pi_T^2) \ll_{\chi} T \cdot \log T$.

Proof. The three terms with π_T : bound $|\pi_T| \ll T^{-\frac{1}{2}}$ on $[T, 2T]$ and $\int_T^{2T} \int_T^{2T} \widehat{\varphi}_T^2(\tau - \tau')^2 \leq T \cdot \int_{\mathbb{R}} \varphi_T^4 \ll T \cdot \log T$; then $|\mu\pi_T| \ll T^{-\frac{1}{2}} \log T$, $|\lambda_T\pi_T| \ll 1$, $|\pi_T^2| \ll T^{-1}$ give respectively $\ll \sqrt{T} \log^2 T$, $\ll T \log T$, $\ll \log T$.

The $2\mu\lambda_T$ term: extend $\tau' \rightarrow \mathbb{R}$ (cost $\ll \log T \cdot \sqrt[3]{T} \cdot (\log T)^{2/3}$), then use the inner-integral formula from the previous Claim:

$$2 \int_T^{2T} d\tau \mu(\tau) \cdot \left(-\frac{1}{\pi} \sum_m \frac{\Lambda(m)}{\sqrt{m}} \cdot g\left(\frac{\log m}{2\pi}\right) \cdot \cos(\tau \log m) \right).$$

Integrate by parts once in τ : $\int_T^{2T} d\tau \mu(\tau) \cdot \cos(\tau \log m) = \left[\frac{\mu(\tau) \sin(\tau \log m)}{\log m} \right]_T^{2T} - \int_T^{2T} d\tau \frac{\mu'(\tau) \sin(\dots)}{\log m} \ll \frac{\log T}{\log m}$ (boundary $\ll \frac{\log T}{\log m}$; integral $\ll \frac{1}{\log m}$ via $\mu' \ll T^{-1}$). Then $\sum_m \frac{\Lambda(m)}{\sqrt{m} \cdot \log m} = \sum_{p \leq T} \frac{1}{\sqrt{p}} + O(1) \ll \frac{\sqrt{T}}{\log T}$; times $\|g\|_\infty \ll \log T$ and the $\frac{\log T}{\log m}$: total $\ll \log^2 T \cdot \frac{\sqrt{T}}{\log T} \ll \sqrt{T} \cdot \log T$. $\square$

Claim. $\left(\frac{N_{\text{all}}(T)}{T} \right)^2 \cdot \int_{\mathbb{R}} \varphi_T^4 + 2 \int_0^{N_{\text{all}}(T)/T} dy y \cdot (\varphi_T^2 * \varphi_T^2)(y) = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} \right) \cdot \frac{N_{\text{all}}(T)}{T} \cdot \left(\int_{\mathbb{R}} \varphi_T^2 \right)^2 + O_\chi(\log^2 T)$.

Proof. Since $\psi'' + 2\psi = 0$ on $(-\frac{1}{2}, \frac{1}{2})$, the map $u \mapsto \psi(u) + \int_{-\frac{1}{2}}^{\frac{1}{2}} dv |u-v| \cdot \psi(v)$ has vanishing second derivative there (differentiate under the integral: $\frac{d^2}{du^2} \int dv |u-v| \cdot \psi(v) = 2\psi(u)$), hence is affine, hence (being even) constant on $(-\frac{1}{2}, \frac{1}{2})$. Its value at $u = 0$ is $\psi(0) + 2 \int_0^{\frac{1}{2}} dv v \cdot \psi(v) = \cos \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \sin \frac{1}{\sqrt{2}}$; and $\int_{-\frac{1}{2}}^{\frac{1}{2}} \psi = \sqrt{2} \sin \frac{1}{\sqrt{2}}$; the ratio is $\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}$. Integrate the constant against ψ over $[-\frac{1}{2}, \frac{1}{2}]$:

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \psi^2 + \int_{-\frac{1}{2}}^{\frac{1}{2}} du \int_{-\frac{1}{2}}^{\frac{1}{2}} dv |u-v| \cdot \psi(u) \cdot \psi(v) = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} \right) \cdot \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \psi \right)^2;$$

the double integral equals $2 \int_0^1 dw w \cdot (\psi * \psi)(w)$ (substitute $w = u - v$, use ψ even).

Rescale: $\varphi_T^2(x)$ and $\psi\left(\frac{xT}{N_{\text{all}}(T)}\right) \cdot \mathbf{1}_{|x| \leq \frac{N_{\text{all}}(T)}{2T}}$ differ only on the two transition intervals of total length $O_\chi(1)$, on which both are ≤ 1 . Hence $\int_{\mathbb{R}} \varphi_T^2 = \frac{N_{\text{all}}(T)}{T} \cdot \int_{-\frac{1}{2}}^{\frac{1}{2}} \psi + O_\chi(1)$ and $\int_{\mathbb{R}} \varphi_T^4 = \frac{N_{\text{all}}(T)}{T} \cdot \int_{-\frac{1}{2}}^{\frac{1}{2}} \psi^2 + O_\chi(1)$; and $(\varphi_T^2 * \varphi_T^2)(y) = \frac{N_{\text{all}}(T)}{T} \cdot (\psi * \psi)\left(\frac{yT}{N_{\text{all}}(T)}\right) + O_\chi(1)$ uniformly, so $\int_0^{N_{\text{all}}(T)/T} dy y \cdot (\varphi_T^2 * \varphi_T^2)(y) = \left(\frac{N_{\text{all}}(T)}{T} \right)^3 \cdot \int_0^1 dw w \cdot (\psi * \psi)(w) + O_\chi(\log^2 T)$. Substitute. $\square$

Assemble: the $\mu\mu$ and $\lambda_T \lambda_T$ main terms sum to T times the left side of the last Claim, so

$$\int_T^{2T} d\tau \int_T^{2T} d\tau' \widehat{\varphi_T^2}(\tau - \tau')^2 \cdot \nu_T \cdot \nu_T = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} \right) \cdot N_{\text{all}}(T) \cdot \left(\int_{\mathbb{R}} \varphi_T^2 \right)^2 + O_\chi(T \cdot \log^2 T).$$

Multiply by $\left(\frac{N_{\text{all}}(T)}{T} \right)^2$, add the first Claim's error $O_\chi(T \cdot \log^4 T \cdot \log \log T)$, and divide through by $\left(\frac{N_{\text{all}}(T)}{T} \cdot \int_{\mathbb{R}} \varphi_T^2 \right)^2 \asymp \log^4 T$: this gives $\|W_T + E_{\text{far}}\|_{\text{HS}}^2 = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} \right) \cdot N_{\text{all}}(T) + O_\chi(T \cdot \log \log T)$. Finally $|\|W_T\|_{\text{HS}}^2 - \|W_T + E_{\text{far}}\|_{\text{HS}}^2| \leq 2 \cdot \|W_T + E_{\text{far}}\|_{\text{HS}} \cdot \|E_{\text{far}}\|_{\text{HS}} + \|E_{\text{far}}\|_{\text{HS}}^2 \ll \sqrt{N_{\text{all}}(T)} \cdot T^{\frac{1}{2}-2\delta}$ by Lemma 3.1. $\square$

Lemma 3.4. Let A and B be Hermitian forms on $\mathbb{C}^n$ with A positive semidefinite. Then:

$$\text{rank } A \geq 2 \cdot \text{tr } A + 4 \cdot \text{tr } B - 4n_+(B) - \|A + B\|_{\text{HS}}^2.$$

Proof of Lemma 3.4. Write $r := \text{rank } A$ and $F(A, B) := \text{rank } A - 2 \cdot \text{tr } A - 4 \cdot \text{tr } B + 4n_+(B) + \|A + B\|_{\text{HS}}^2$. The minima of $U \rightarrow \mathbb{R}$ via $U \mapsto F(A, U \cdot B \cdot U^\dagger) =: F(A, B_U)$ satisfy $\text{tr}(A \cdot [B_U, H]) = 0 = \text{tr}([A, B_U] \cdot H)$ for all Hermitian H , whence at a minimum $[A, B_U] = 0$. But then for the purpose of the inequality we may assume without loss of generality that A and B are diagonal, with eigenvalues $a_1 \geq \dots \geq a_n \geq 0$ (so $a_i = 0$ for $i > r$) and $b_1, \dots, b_n$. Then with $x_i := a_i + b_i$,

$$\|A + B\|_{\text{HS}}^2 = \sum_{i=1}^n x_i^2 \geq \sum_{i=1}^n 2a_i + 4b_i - c_i = 2 \text{tr } A + 4 \text{tr } B - \sum_i c_i,$$

where $c_i = 1, 0, 4$ according as $i \leq r, b_i \leq 0$; $i > r, b_i \leq 0$; $b_i > 0$ (via $(x_i - 1)^2, x_i^2, (x_i - 2)^2 \geq 0$ and $-2b_i, -4b_i, 2a_i \geq 0$ respectively); and $\sum_{i=1}^n c_i \leq r + 4 \cdot \#\{i : b_i > 0\} = r + 4n_+(B)$. Rearranging we conclude. $\square$